Team #: 3

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WORKSHOP: Network Analysis – Part I (NodeXL)

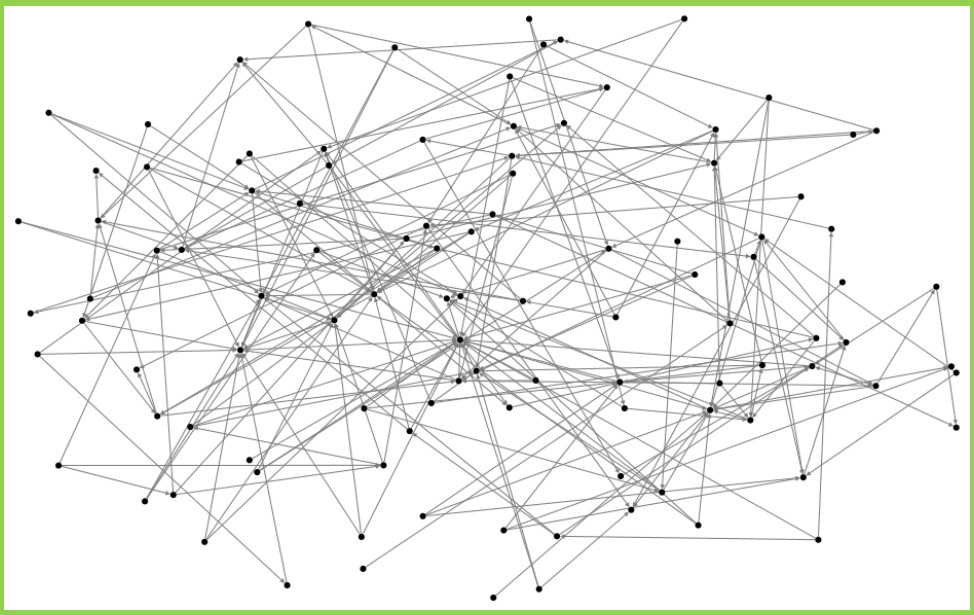
Objective: Using an Excel plugin NodeXL visualize and analyze structure of a small online brand community. Learn how to apply this tool to derive key metrics used in Social Network Analysis (SNA). Explore applications of network analysis in marketing decision making.

Analysis Plan

Step 1. Data Preparation

(a) Launch MS-EXCEL and create new workbook using NodeXLGraph template.
(b) Copy and paste network data from "brand_sna.txt" file posted on the class website to the "Edges" worksheet. (In case, the data is being pasted as a single column, use "Data" -> "Text to Columns" -> "Delimited" -> "Space" option in EXCEL to split it into two columns)
(c) Visualize network by clicking "Show/Refresh Graph" button. You should see a network of 231 edges and 106 vertices.
(d) Switch between "directed" and "undirected" graph options and refresh the graph. Note the difference.
(e) **Hand in**: Picture of "directed" graph

< INSERT PICTURE OF "DIRECTED" GRAPH HERE>



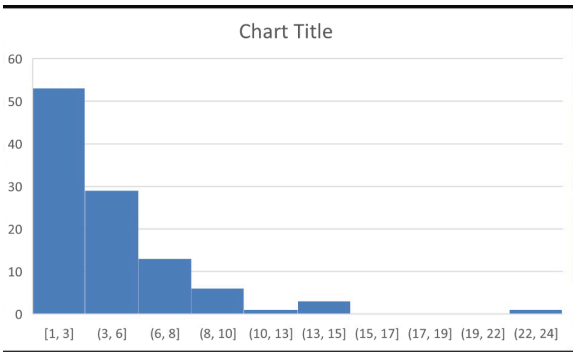
Step 2. Generate Graph Metrics

(a) Switch to “Undirected” network view.
(b) Click “Graph metrics”, select “All”, and click “Calculate Metrics”.
(c) Look at the “Overall Metrics” worksheet. Make sure you understand what each metric represents (if necessary, refer to Wikipedia or search other online resources).
(d) **Hand in**: The following summary metrics: Maximum and Average Geodesic Distance, Graph Density, Minimum, Maximum, Average, Median Degree. **Explain** what is captured by each of these metrics.

Metric	Value	Captures (Description)
Maximum Geodesic Distance	7	The longest shortest-path distance between any two vertices in the network; shows the network's overall span.
Average Geodesic Distance	3.252403	The average number of steps in the shortest paths between all pairs of vertices; shows how easily people can reach out another on average.
Graph Density	0.041509434	The proportion of actual connections compared with all possible connections in the network; shows how tightly connected the overall network is.
Minimum Degree	1	The smallest number of direct connections held by any vertex shows the least-connected individual in the network.
Maximum Degree	23	The largest number of direct connections held by any vertex shows the most-connected.
Average Degree	4.358	The mean number of direct connections per vertex shows the typical connectivity level in the network.
Median Degree	3.500	The middle degree value when all vertices are ordered from lowest to highest degree shows the central tendency of connectivity while being less affected by extreme values.

(e) **Hand in**: Degree distribution (histogram) and your comment.

< INSERT DEGREE DISTRIBUTION (HISTOGRAM) HERE>



Comment:
The more Y → smaller X
A smaller X → a bigger Y

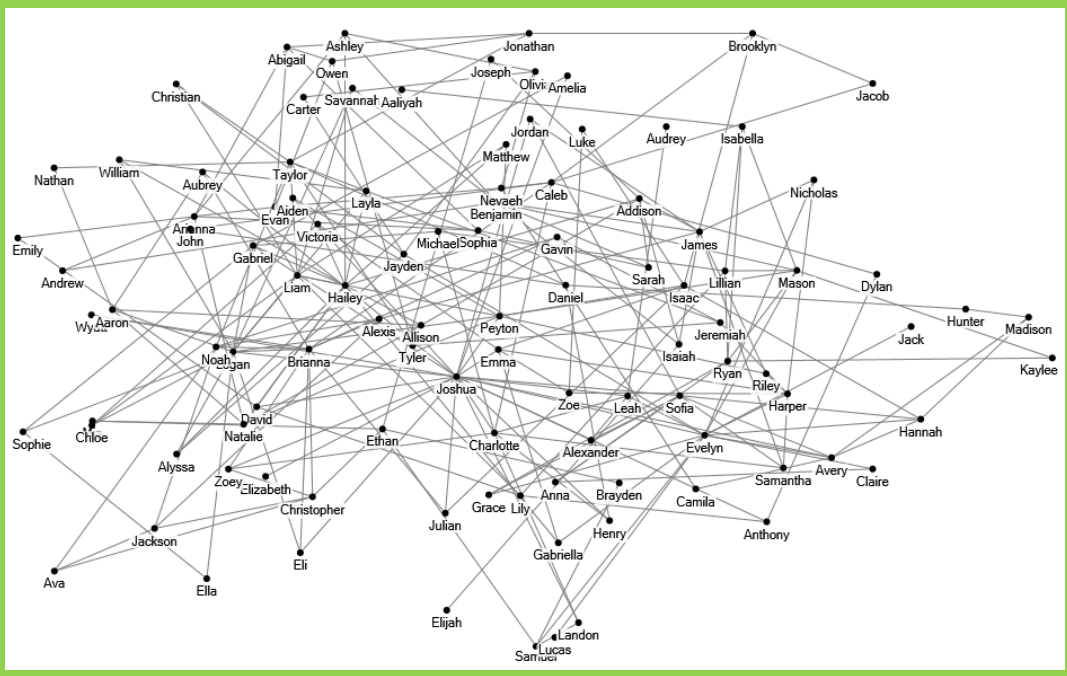
(f) Look at “Vertices” worksheet under “Graph Metrics” columns. Make sure you understand what each metric represents.

Step 3. Network Visualization

(a) Using autofill option map Vertex Degree to “Vertex Size” (1 - 100), Betweenness Centrality (40 - 100) - to “Vertex Opacity”, Closeness Centrality set as the “Vertex Tooltip” and Vertex to “Vertex Label”.

(b) **Hand in**: Network visualization.

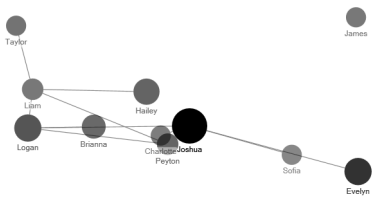
< INSERT PICTURE OF THE NETWORK HERE >



(c) Explore “Dynamic filters” option. Set filters for “Degree” using different cutoff values. (You may need to click “Lay Out Again” button a few times to find optimal visualization.)

(d) **Hand in**: Set “Degree” filter to “8 and higher” and copy and paste network visualization to your report. Comment if your filtered network is fully connected or has disconnected elements.

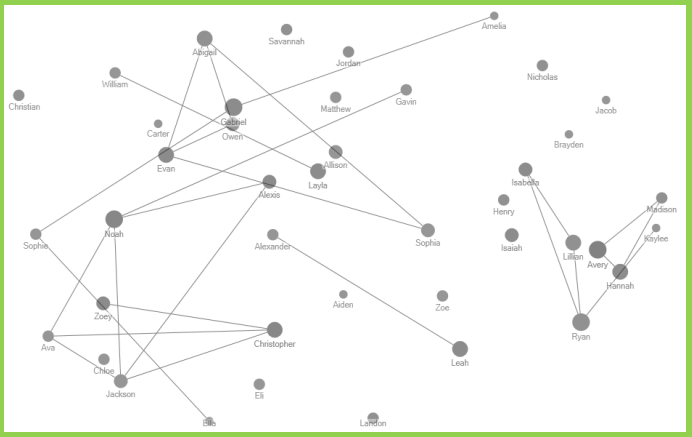
< INSERT PICTURE OF THE NETWORK HERE >



Comment: There is one disconnected element : James.

(e) **Hand in**: Reset filters. Set “Clustering coefficient” filter to “0.2 and higher”. Copy and paste network visualization to your report. Comment on how many disconnected components you observe.

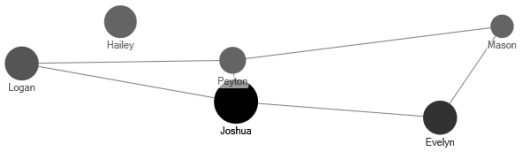
< INSERT PICTURE OF THE NETWORK HERE >



Comment:
11 components are completely disconnected from every other cluster. There are two clusters that are disconnected from the major cluster, but still have in-betweeness within their own groups.

(f) **Hand in**: Reset filters. Set “Betweenness centrality” filter to “450 and higher”. Copy and paste network visualization to your report. Comment who has the highest Betweenness overall? Who has the highest Betweenness and is not connected to other high Betweenness vertices?

< INSERT PICTURE OF THE NETWORK HERE >

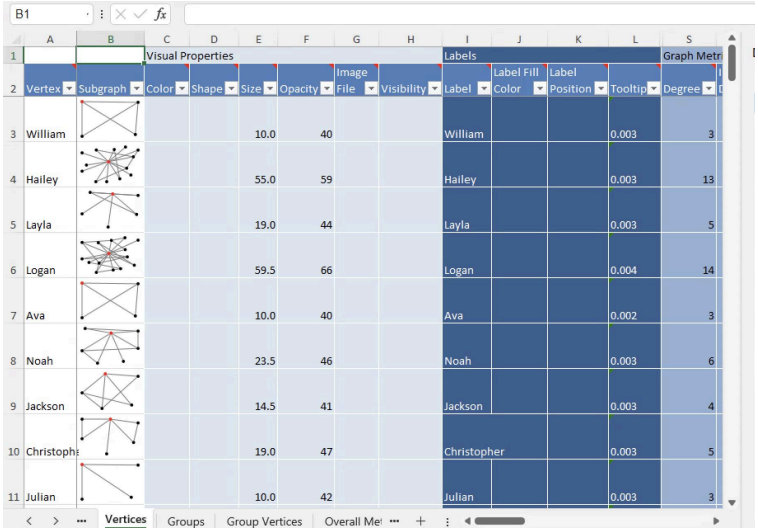


Who has the highest Betweenness overall?
Joshua

Who has the highest Betweenness and is not connected to other high Betweenness vertices? Hailey

(g) Reset filters.

(h) Generate Subgraph Images using level 1.5. Explore subgraphs added to worksheet “Vertices”.

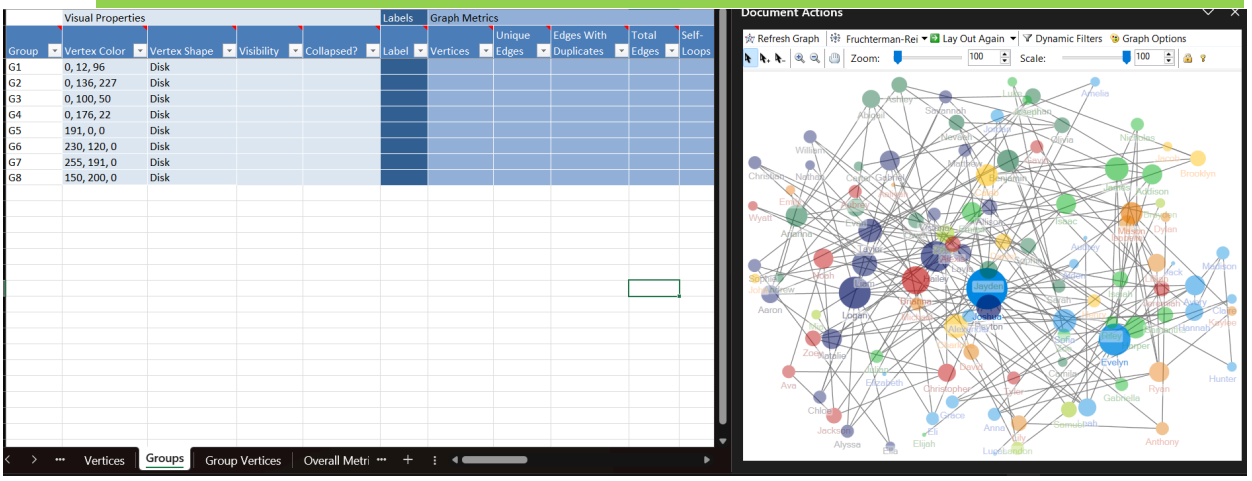


	Vertex	Subgraph	Color	Shape	Size	Opacity	Image File	Visibility	Label	Label Fill Color	Label Position	Tooltip	Degree
3	William				10.0	40			William			0.003	3
4	Hailey				55.0	59			Hailey			0.003	13
5	Layla				19.0	44			Layla			0.003	5
6	Logan				59.5	66			Logan			0.004	14
7	Ava				10.0	40			Ava			0.002	3
8	Noah				23.5	46			Noah			0.003	6
9	Jackson				14.5	41			Jackson			0.003	4
10	Christoph				19.0	47			Christopher			0.003	5
11	Julian				10.0	42			Julian			0.003	3

Step 4. Explore Groups

(a) **Hand in**: Group vertices by Cluster using Clauset-Newman-Moore algorithm (under “Groups” menu). Collapse all groups. Click “Refresh Graph”. How many groups were created? Copy and paste network visualization to your report.

< INSERT PICTURE OF THE NETWORK HERE >



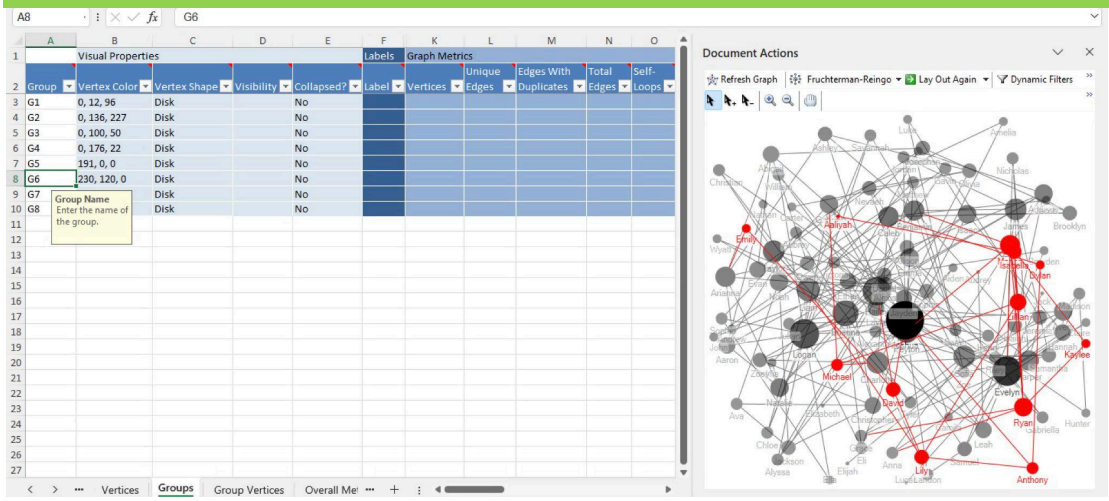
Group	Vertex Color	Vertex Shape	Visibility	Collapsed?	Label	Vertices	Unique Edges	Edges With Duplicates	Total Edges	Self-Loops
G1	0, 12, 96	Disk								
G2	0, 136, 227	Disk								
G3	0, 100, 50	Disk								
G4	0, 176, 22	Disk								
G5	191, 0, 0	Disk								
G6	230, 120, 0	Disk								
G7	255, 191, 0	Disk								
G8	150, 200, 0	Disk								

How many groups were created? 8 groups were created

(b) Using “Group Vertices” worksheet you can look up group membership for a given person. Switching to “Groups” worksheet, you can highlight and select the corresponding group on the graph. Next, by choosing NodeXL Menu “Groups” -> “Expand Selected Groups”, you can inspect within and (possibly) outside the group connections of the selected individual.

(c) Identify and expand the group for David (keep other groups collapsed). Copy and paste network visualization of the David connections to your report.

< INSERT PICTURE OF THE NETWORK HERE >



(d) Using information from "Group Vertices" and "Edges" worksheets check if David has any connections *outside* his group. (*Hint*: consider using EXCEL **vlookup** function)

Does David have any connections outside his group? If yes, list these here and provide corresponding group labels.

Vertex 1	Vertex 2
David	Emily
David	Hailey
David	Joshua

(Emily is in group 6 as well, so would not be considered a connection outside of his group)

Vertex	Group Label
Joshua	2
Hailey	1
…	…

Step 5. Exploring Targeting Strategies

(a) Assume that you need to select only one person in this network to target your viral marketing campaign to. Who would be that person? *Explain your choice.*

Your answer here:

Joshua due to him being the individual with the highest betweenness amongst the group. He also has the largest network based off of the subgraph we created earlier on.

(b) Create two lists of top ten individuals based on (i) betweenness and (ii) degree centrality. Are you getting the same people in both lists? What would be your considerations when choosing which list to use for targeting?

Rank	Top by Betweenness	Top by Degree Centrality
1	Joshua	Joshua
2	Evelyn	Evelyn
3	Logan	Logan
4	Peyton	Hailey
5	Mason	Brianna
6	Hailey	Peyton
7	Brianna	Liam
8	Taylor	Taylor
9	Calen	James
10	Benjamin	Charlotte

Are you getting the same people in both lists?

The first three individuals are the same on both lists, but there is more variance after these three.

What would be your considerations when choosing which list to use for targeting?

We would consider Joshua due to his holding the top spot for having the largest in-betweenness and most degreed individual. As well as he did have the largest network based off of our subgraph. These are a few reasons as to why we should consider Joshua for our viral Marketing campaign.

(c) For each group identify the most connected (highest degree) individual. Under what conditions it might be more effective to target group level “influencers” versus the overall top connected individuals (as in the previous question)?

Group	Highest degree individual
1	Hailey
2	Joshua
3	Benjamin
4	James
5	Brianna
6	Caleb
7	Charlotte
8	Ethan

(add rows as needed)

Under what conditions it might be more effective to target group level "influencers" versus the overall top connected individuals (as in the previous question)?

It would only be beneficial to target group level influencers when clusters are loosely connected, or if the individuals with the highest connections are all within one group making it hard to reach other individuals. Targeting overall connected individuals would be more efficient in this case since the clusters all mostly have strong connections and messaging will travel through groups naturally.

(d) Abigail and Layla both have a degree of 5. Who would you target? *Explain.* (Hint: refer to your analysis in Step 4)

Who would you target?

Layla

Explanation:

While Abigail has a larger network based on the subgraphs that we created, we would choose to target Layla. Her network is a bit smaller, but considering that the difference in connections between Groups 1 and 3 are so noticeable, and group 3 has less individuals representing them with higher levels of degrees or in-betweenness, Layla would be a better target to spread the word amongst her group. Many individuals in group 1 have high levels of in-betweenness and degrees, so the message will travel through the network on its own.

(e) When Ethan and Leah discover a special offer from your company, whose friends (i.e. direct contacts) are more likely to learn about the offer quicker? *Explain.* (Hint: refer to your analysis in step 3.h)

Whose friends are more likely to learn about the offer quicker?

Leah.

Leah				19.0	45			Leah			0.003	5			127.129
Ethan				19.0	46			Ethan			0.003	5			134.726

Explanation:

Since the two both have a degree of 5, their friends would likely find out about the offer around the same time if they had similar networks of friends. As we can see by looking at Leah's subgraph, there are more connections between the individuals in her network so, intuitively, even if Leah does not tell one person immediately there is more opportunity for word to travel quicker than for Ethan's network where he is the only mutual connection between all individuals in his network.